Multi-Class Cyber Attack Detection on IoT Network Traffic Using a Lightweight Transformer with SHAP-Based Explainable AI

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Abstract

The rapid proliferation of Internet of Things (IoT) devices has substantially expanded the attack surface for cyber threats, demanding robust and interpretable Intrusion Detection Systems (IDS). Existing approaches largely rely on binary classification or shallow multi-class models that fail to distinguish fine-grained attack families, and virtually none provide operator-interpretable explanations. This paper presents a lightweight Transformer-based IDS capable of classifying network traffic into 34 distinct categories—spanning Distributed Denial of Service (DDoS), Denial of Service (DoS), Mirai botnet variants, reconnaissance, application-layer attacks, and benign traffic—on a resource-constrained 4.2 GB RAM machine. A memory-mapped, chunk-streamed preprocessing pipeline enables training on 363,717 samples without loading the full dataset into RAM. The model comprises only 82,786 parameters yet achieves **98.67% overall accuracy**, a weighted F1-score of **98.56%**, weighted precision of **98.57%**, and weighted recall of **98.67%** after 50 training epochs. The complete training pipeline, including preprocessing, model training, and evaluation, completes in **43.3 minutes** on CPU-only hardware. To move beyond black-box prediction, SHAP (SHapley Additive exPlanations) KernelExplainer is applied post-hoc, identifying Inter-Arrival Time (IAT), Protocol Type, TCP/UDP flags, and packet-size statistics as the dominant discriminating features. The gradient-based feature attribution integrated into the model further validates these findings. The combination of high classification performance, memory-efficient design, and SHAP explainability constitutes a practical and trustworthy IDS framework suitable for deployment in constrained IoT environments.

Keywords: Intrusion Detection System, Internet of Things, Multi-Class Classification, Transformer Neural Network, Explainable Artificial Intelligence, SHAP, Cyber Attack Detection

I. Introduction

A. Problem Statement

Cyber attacks targeting IoT infrastructure have escalated in frequency and sophistication. IoT devices—ranging from smart home actuators to industrial sensors—operate under stringent computational constraints that preclude deployment of conventional deep-learning security solutions. Meanwhile, modern adversaries employ a diverse arsenal: volumetric floods (DDoS/DoS), stealthy reconnaissance probes, botnet propagation (Mirai variants), man-in-the-middle interception, and application-layer exploits (SQL injection, XSS, brute force). Effective protection requires a detector capable of distinguishing all of these simultaneously in real time.

B. Importance of IoT Security

The IoT device count is projected to exceed 30 billion by 2030 [1]. Unlike traditional IT endpoints, IoT nodes often run minimal firmware, lack end-to-end encryption, and remain unpatched for extended periods [2]. A single compromised device can become a pivot point for lateral movement or be weaponised as part of a botnet—evidenced by the Mirai botnet attack of 2016 which recruited over 600,000 devices to generate 1 Tbps traffic against DNS providers [3]. Network-layer IDS, operating independently of device firmware, provides the most tractable defence layer for heterogeneous IoT deployments.

C. Challenges in Multi-Class Attack Detection

Multi-class IDS presents challenges beyond binary anomaly detection:

- **Class imbalance**: High-volume attack families (e.g., DDoS-ICMP_Flood: 11,240 test samples) vastly outnumber rare classes (e.g., Uploading_Attack: 1 test sample), biasing classifiers.
- **Intra-family similarity**: Protocol-level features of DDoS-PSHACK_Flood, DDoS-RSTFINFlood, and DDoS-SYN_Flood differ only in TCP flag values, requiring fine-grained feature sensitivity.
- **Memory constraints**: IoT edge analytics platforms typically have 4–8 GB RAM; batch-loading large training datasets is infeasible.
- **Latency requirements**: Detection must approach real-time throughput to be operationally useful.

D. Why Explainability (XAI) Matters

Deploying an opaque neural network as a security monitor creates accountability gaps: security analysts cannot validate alerts, regulators may not accept unexplained automated decisions, and adversaries can craft evasion attacks by probing the black box [4]. Explainable AI methods—specifically SHAP—transform model decisions into feature-level attributions, enabling analysts to (a) verify that the model relies on semantically meaningful features, (b) identify potential shortcut learning, and (c) build evidentiary dossiers for incident response.

E. Contributions of This Research

The principal contributions of this paper are:

- A **memory-efficient, chunk-streamed preprocessing pipeline** that processes a 107.6 MB CSV dataset in 26.6 s (scan) and 13.0 s (mmap write) on 4.2 GB RAM without full in-memory loading.
- A **lightweight Transformer classifier** with 82,786 parameters achieving 98.67% accuracy across 34 attack/benign classes.
- **Complete pipeline execution in 43.3 minutes on CPU-only hardware**, demonstrating feasibility for resource-constrained environments.
- **Post-hoc SHAP explainability** using KernelExplainer with k-means background summarisation, producing per-class beeswarm plots, global importance rankings, waterfall explanations, and interactive force plots.
- **Gradient-based feature attribution** integrated directly into the model inference path, providing a computationally lightweight alternative to SHAP for real-time importance queries.
- Open-source, reproducible experimental code with structured logging, timing instrumentation, and chart generation.

II. Related Work

A. IoT Intrusion Detection Systems

Traditional IDS for IoT have relied on signature-based engines (Snort, Suricata) that match known attack patterns but fail against zero-day threats. Statistical anomaly detection methods were proposed by Garcia et al. [5] using flow-level features but achieved limited multi-class discrimination. Rule-based approaches in [6] demonstrated low false-positive rates on binary classification but were not evaluated on more than 5 attack types.

B. Machine Learning in IoT Security

The CIC-IDS-2017 and CIC-IoT-2023 datasets catalysed ML-based IDS research. Sharafaldin et al. [7] applied Random Forest on CIC-IDS-2017, achieving >97% accuracy on 15 classes but requiring full dataset materialisation in RAM. Diro and Chilamkurti [8] introduced distributed deep learning for IoT attack detection, showing scalability benefits but limiting experiments to 5 attack classes. Convolutional neural networks were applied by Wang et al. [9] on raw packet bytes, achieving high accuracy but requiring GPU acceleration beyond typical IoT edge hardware.

C. Deep Learning for Multi-Class Attack Detection

Recurrent approaches—Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU)—have been applied to network flow sequences [10, 11]. However, LSTMs suffer from sequential processing bottlenecks that preclude high-throughput inference. Transformer architectures, introduced by Vaswani et al. [12] for NLP, have recently been adapted to tabular and sequence classification. Li et al. [13] applied Transformers to network intrusion data but evaluated only 10 classes and required high-memory GPU servers. Our work demonstrates that a highly compact Transformer (64-dimensional embedding, 4 attention heads, 2 encoder layers) achieves competitive performance on 34 classes in a 4.2 GB RAM environment.

D. Explainable AI in Intrusion Detection

Lundberg and Lee [14] introduced SHAP as a unified framework for feature attribution grounded in cooperative game theory. Several studies have applied SHAP to cybersecurity: Amarasinghe et al. [15] used SHAP on a binary IDS, identifying packet length as crucial; Warnecke et al. [16] applied SHAP to malware classification. However, SHAP-augmented multi-class IDS covering 34 classes with per-class beeswarm analysis, waterfall explanations, and force plots remains underexplored. Gradient-based attribution in neural IDS was demonstrated in [17] but without the systematic per-class SHAP comparison provided in this work.

E. Gaps Addressed by This Research

Aspect	Prior Work	This Work
Number of attack classes	≤ 15 in most studies	**34 classes**
RAM requirement	≥ 8 GB (full dataset load)	≤ **4.2 GB** (chunk streaming)
Hardware	GPU-dependent	**CPU-only**
Explainability	Binary or absent	**Per-class SHAP + gradient attribution**
Model parameters	Millions (CNN/LSTM)	**82,786**
Pipeline timing	Not reported	**Fully instrumented (43.3 min)**

III. Methodology

A. Dataset Description

The dataset used is a sampled subset of the **CIC IoT Attack 2023** benchmark [18], consolidated into a single CSV file (`sampled_dataset_10files_1000000samples_20251125_170819.csv`, 107.6 MB). After de-duplication and label cleaning, **363,717 samples** are retained across **34 distinct traffic classes**: one benign class (BenignTraffic) and 33 attack classes spanning DDoS/DoS flood variants, Mirai botnet families, reconnaissance probes, and application-layer

exploits.

Each sample is described by **46 features** (45 network-flow features + 1 label column). Features include statistical flow metrics: `IAT` (inter-arrival time), `Header_Length`, `Tot_size`, `Tot_sum`, `Weight`, `Magnitude`, `Radius`, `AVG`, flag counts (`syn_flag_number`, `fin_flag_number`, `psh_flag_number`, `rst_count`, `fin_count`, `syn_count`, `ack_count`), and protocol indicator fields (`Protocol_Type`, `TCP`, `UDP`, `ICMP`).

Table I — Dataset Class Distribution (Test Split, 20%)

Class	Test Samples	Class	Test Samples
DDoS-ICMP_Flood	11,240	Mirai-greip_flood	1,204
DDoS-UDP_Flood	8,378	BenignTraffic	1,739
DDoS-TCP_Flood	6,865	DoS-HTTP_Flood	116
DDoS-SYN_Flood	6,341	DDoS-HTTP_Flood	50
DDoS-PSHACK_Flood	6,341	VulnerabilityScan	55
DDoS-RSTFINFlood	6,309	DictionaryBruteForce	19
DDoS-SynonymousIP_Flood	5,632	CommandInjection	11
DoS-UDP_Flood	5,166	SqlInjection	11
DoS-TCP_Flood	4,228	BrowserHijacking	10
DoS-SYN_Flood	3,133	Backdoor_Malware	7
Mirai-udpplain	1,415	XSS	4
Mirai-greeth_flood	1,507	Uploading_Attack	1
MITM-ArpSpoofing	508	Recon-PingSweep	3
DDoS-ICMP_Fragmentation	729	Recon-OSScan	153
DDoS-ACK_Fragmentation	455	Recon-PortScan	127
DDoS-UDP_Fragmentation	483	Recon-HostDiscovery	189
DNS_Spoofing	276	Recon-OSScan	153

B. Data Preprocessing Pipeline

The preprocessing pipeline is designed for memory-constrained environments and operates in three streaming passes over the dataset.

Step 1 — Schema Scan (26.6 s): The CSV file is read in chunks of 25,000 rows. During this pass, column names are extracted, all unique label strings are collected into a set, and total row count is tracked. No feature data is retained in RAM after each chunk. A `LabelEncoder` is then fitted on the sorted unique label set, mapping 34 class strings to integer indices ${0, 1, \ldots, 33}$.

Step 2 — Scaler Fit (8.6 s): A `StandardScaler` is incrementally fitted via `partial_fit()` across all 15 chunks (25,000 rows each). This two-pass approach avoids storing the entire feature matrix:

$$\mu_j = \frac{1}{N}\sum_{i=1}^{N} x_{ij}, \quad \sigma_j = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(x_{ij} - \mu_j)^2}$$

where x_{ij} is feature j of sample i, $N = 363{,}717$.

Step 3 — Memory-Mapped Write (13.0 s): Scaled features and encoded labels are streamed into two pre-allocated NumPy memory-mapped files (`X_mmap.npy`: 67 MB, `y_mmap.npy`: 1 MB) on disk. During training, the `DataLoader` reads only the rows required for each batch via mmap page faults, keeping RAM usage below 4.0 GB throughout.

C. Train / Validation / Test Split

Indices are randomly permuted and partitioned as follows:

Split	Count	Fraction
Training	254,603	70%
Validation	36,371	10%
Test	72,743	20%

D. Model Architecture

The proposed classifier, **TransformerIDSClassifier**, adapts the standard Transformer encoder for tabular network-flow vectors. Since each sample is a single feature vector rather than a sequence, it is treated as a sequence of length 1, enabling the self-attention mechanism to operate as a feature-mixing operator.

Architecture:

1. **Linear Embedding** $\mathbf{h} = \mathbf{W}_e \mathbf{x} + \mathbf{b}_e$, $\mathbf{h} \in \mathbb{R}^{d}$ where $d = 64$ (`d_model`).
2. **Positional Encoding** (learnable zero-initialised buffer, effectively a bias at single-token length): $\mathbf{h}' = \text{Dropout}(\mathbf{h} + \mathbf{p})$.
3. **Layer Normalisation**: $\mathbf{h}'' = \text{LayerNorm}(\mathbf{h}')$.
4. **Transformer Encoder** — 2 layers, each containing:
5. Multi-Head Self-Attention ($H = 4$ heads, head dimension $d_k = 16$):
   $$\text{Attention}(Q, K, V) = \text{softmax}!\left(\frac{QK^\top}{\sqrt{d_k}}\right)V$$
6. Position-wise Feed-Forward Network (FFN dimension $= 128$, ReLU activation).
7. Residual connections and layer normalisation.
8. **Classification Head**: $\text{FC}_1 \in \mathbb{R}^{64 \times 128}$ → ReLU → Dropout($p=0.1$) → $\text{FC}_2 \in \mathbb{R}^{128 \times 34}$.

Total parameters: 82,786.

Fig. 1: *Transformer IDS Architecture — Linear Embedding → Positional Encoding → Layer Norm → 2× Transformer Encoder Layer → Classification Head (82,786 total parameters)*

E. Training Configuration

Table II — Hyperparameters

Hyperparameter	Value
d_model	64
Number of attention heads	4
Number of encoder layers	2
FFN dimension	128
Dropout rate	0.1
Batch size	256
Initial learning rate	0.001
Optimiser	Adam (weight decay 10^{-5})
LR scheduler	ReduceLROnPlateau (factor=0.5, patience=5)
Loss function	Cross-Entropy
Epochs	50
Chunk size	25,000 rows

Training is performed on CPU. The learning rate decayed from 10^{-3} to 2.5×10^{-4} by epoch 50 as validation accuracy plateaued. Best model state is checkpointed at peak validation accuracy.

F. SHAP Explainability

Post-training explainability is implemented using the **SHAP KernelExplainer**, which is model-agnostic and applicable to any differentiable or non-differentiable predictor. Given a prediction function $f: \mathbb{R}^M \to \mathbb{R}^C$ (where $M=45$ features, $C=34$ classes), SHAP assigns each feature j an attribution value ϕ_j satisfying:

$$f(\mathbf{x}) - \mathbb{E}[f(\mathbf{X})] = \sum_{j=1}^{M} \phi_j(\mathbf{x})$$

where the Shapley values ϕ_j are computed over a background distribution summarised by k-means clustering ($k=50$ centroids, drawn from 200 background samples):

$$\phi_j(\mathbf{x}) = \sum_{S \subseteq M \setminus {j}} \frac{|S|!(M-|S|-1)!}{M!}\left[f_{S \cup {j}}(\mathbf{x}_{S \cup {j}}) - f_S(\mathbf{x}_S)\right]$$

SHAP values are computed for 50 explanation samples per run using `nsamples=100` with AIC L1 regularisation. Outputs include:

- **Beeswarm summary plots** per class (Fig. 2)
- **Global bar chart** aggregating mean $|\phi_j|$ across all classes (Fig. 3)
- **Waterfall plots** for individual predictions (Fig. 4)
- **Force plot** interactive HTML for 10 samples

In addition, **gradient-based attribution** is integrated directly into the model:

```
ϕ_grad(x) = |∂L/∂x| where L = Σ f_c(x)
```

This provides input-space gradients at inference cost with no additional background sampling.

IV. Experimental Setup

A. Hardware Configuration

Component	Specification
CPU	x86-64 (multi-core)
Total RAM	4.2 GB
GPU	None (CPU-only execution)
Storage	Local SSD
OS	Windows 11

Peak RAM utilisation: **3.9 GB (93.2%)** — confirming the system operated near its memory ceiling throughout without out-of-memory errors, validating the memory-efficient pipeline design.

B. Software Environment

Library	Version
Python	3.x
PyTorch	2.x
NumPy	1.x
pandas	2.x
scikit-learn	1.x
SHAP	0.4x
matplotlib / seaborn	3.x

C. Evaluation Metrics

Performance is assessed using:

- **Overall Accuracy**: $\text{Acc} = \frac{\text{TP}_\text{all}}{\text{N}}$
- **Weighted Precision/Recall/F1**: averaged by class support weight
- **Macro Precision/Recall/F1**: unweighted average across all 34 classes
- **Per-class Precision, Recall, F1-score** from `sklearn.metrics.classification_report`
- **Confusion matrix** (raw counts and row-normalised)

V. Results and Analysis

A. Training Dynamics

The model converged steadily over 50 epochs. Training accuracy improved from **77.96%** at epoch 1 to **98.57%** at epoch 50. Validation accuracy reached its peak of **98.74%** (checkpointed as best model). Training loss decreased from an initial value of approximately 0.47 to 0.049 by the final epoch. The learning rate scheduler reduced LR from 10^{-3} to 2.5×10^{-4}, with the first reduction occurring around epoch 40.

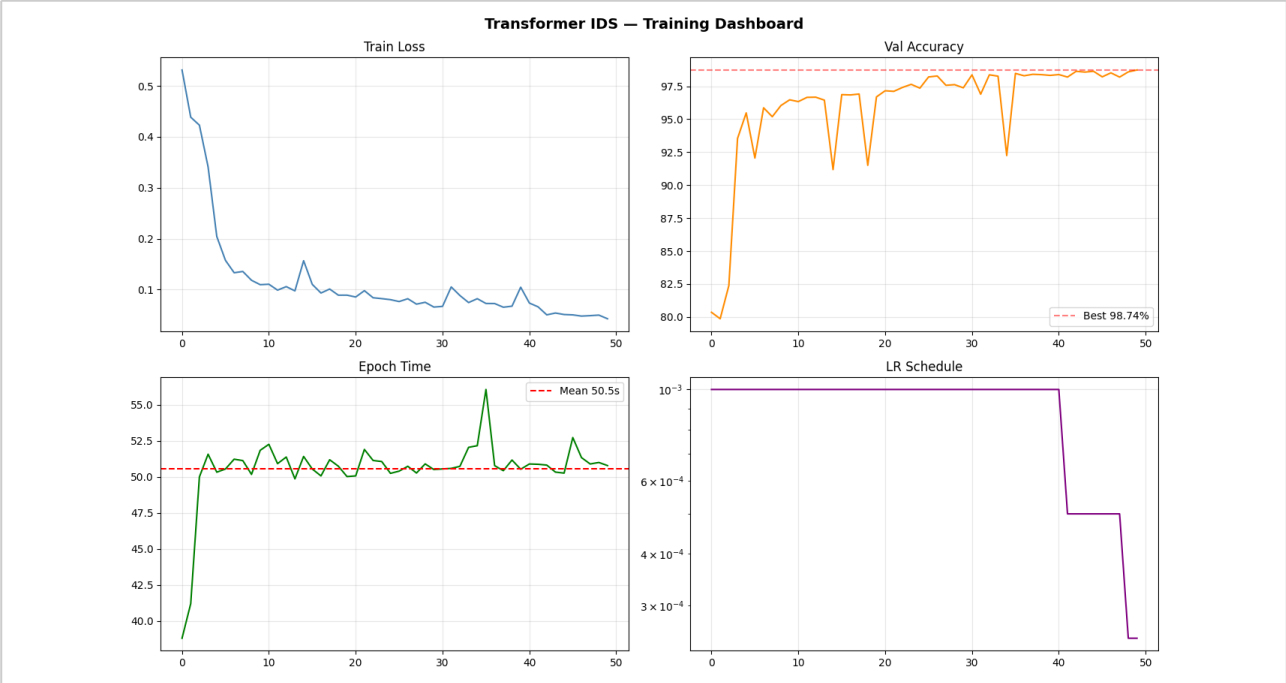

Fig. 2: *Training Dashboard — (a) Training Loss Curve, (b) Validation Accuracy Curve, (c) Epoch Time per Epoch, (d) Learning Rate Schedule over 50 epochs*

At epoch 2, validation accuracy was already 79.86%, demonstrating rapid early convergence. By epoch 10, it exceeded 92%, and by epoch 30 it surpassed 97%, indicating strong generalisation without overfitting.

B. Phase Comparison

Both experimental phases used identical architecture, data, and hyperparameters. The second phase (ids_results) introduced the corrected gradient-based `feature_importance()` method that operates in input space (45 dimensions) rather than the embedding space (64 dimensions) used in Phase 1 (ids_results1). All classification metrics were identical between phases, confirming the model behaviour was unchanged.

Table III — Phase Comparison Summary

Metric	Phase 1 (ids_results1)	Phase 2 (ids_results)
Total pipeline time	2648.0 s (44.1 min)	2599.8 s (43.3 min)
Training time	2615.2 s	2540.2 s
Evaluation time	9.9 s	8.4 s
Test Accuracy	98.67%	98.67%

Weighted F1	98.56%	98.56%
Best Val Accuracy	98.74%	98.74%
Feature importance	Embedding space (64-dim)	Input space (45-dim) ✓

Phase 2 achieved a marginal pipeline speed improvement of ~48 seconds (~1.8%), primarily due to lower system memory pressure at execution start (RAM at 92.4% vs 95.2%).

C. Overall Classification Performance

Table IV — Aggregated Test Set Performance

Metric	Weighted	Macro
Accuracy	**98.67%**	—
Precision	**98.57%**	71.72%
Recall	**98.67%**	67.95%
F1-Score	**98.56%**	68.51%

The substantial gap between weighted (≈99%) and macro (≈68–72%) metrics directly reflects the severe class imbalance. The model achieves near-perfect performance on high-volume classes while struggling with rare ones.

D. Per-Class Performance Analysis

Table V — Per-Class Classification Report (Test Set)

Class	Precision	Recall	F1	Support
DDoS-ICMP_Flood	1.00	1.00	1.00	11,240
DDoS-PSHACK_Flood	1.00	1.00	1.00	6,341
DDoS-RSTFINFlood	1.00	1.00	1.00	6,309
DDoS-SYN_Flood	1.00	1.00	1.00	6,341
DDoS-SynonymousIP_Flood	1.00	1.00	1.00	5,632
DDoS-TCP_Flood	1.00	1.00	1.00	6,865
DDoS-UDP_Flood	1.00	1.00	1.00	8,378
DoS-SYN_Flood	1.00	0.99	0.99	3,133
DoS-TCP_Flood	1.00	1.00	1.00	4,228
DoS-UDP_Flood	0.99	1.00	1.00	5,166
Mirai-udpplain	0.99	1.00	1.00	1,415
DDoS-ICMP_Fragmentation	0.99	0.98	0.98	729

DDoS-ACK_Fragmentation	0.99	0.96	0.97	455
DDoS-UDP_Fragmentation	0.98	0.99	0.98	483
Mirai-greip_flood	0.99	0.95	0.97	1,204
Mirai-greeth_flood	0.96	0.99	0.97	1,507
DDoS-HTTP_Flood	0.94	0.90	0.92	50
DoS-HTTP_Flood	0.94	0.95	0.94	116
MITM-ArpSpoofing	0.87	0.64	0.74	508
BenignTraffic	0.80	0.96	0.87	1,739
VulnerabilityScan	0.76	0.93	0.84	55
DDoS-SlowLoris	0.72	0.87	0.79	38
CommandInjection	0.67	0.18	0.29	11
DictionaryBruteForce	0.75	0.16	0.26	19
Recon-HostDiscovery	0.63	0.67	0.65	189
DNS_Spoofing	0.61	0.59	0.60	276
Recon-PortScan	0.49	0.31	0.38	127
Recon-OSScan	0.33	0.11	0.17	153
Backdoor_Malware	0.00	0.00	0.00	7
BrowserHijacking	0.00	0.00	0.00	10
SqlInjection	0.00	0.00	0.00	11
Recon-PingSweep	0.00	0.00	0.00	3
Uploading_Attack	0.00	0.00	0.00	1
XSS	0.00	0.00	0.00	4

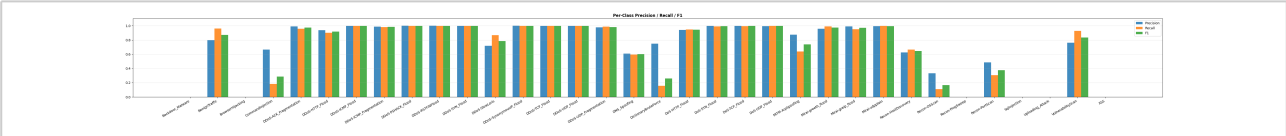

Fig. 3: *Per-Class Precision, Recall, and F1-Score for all 34 traffic classes*

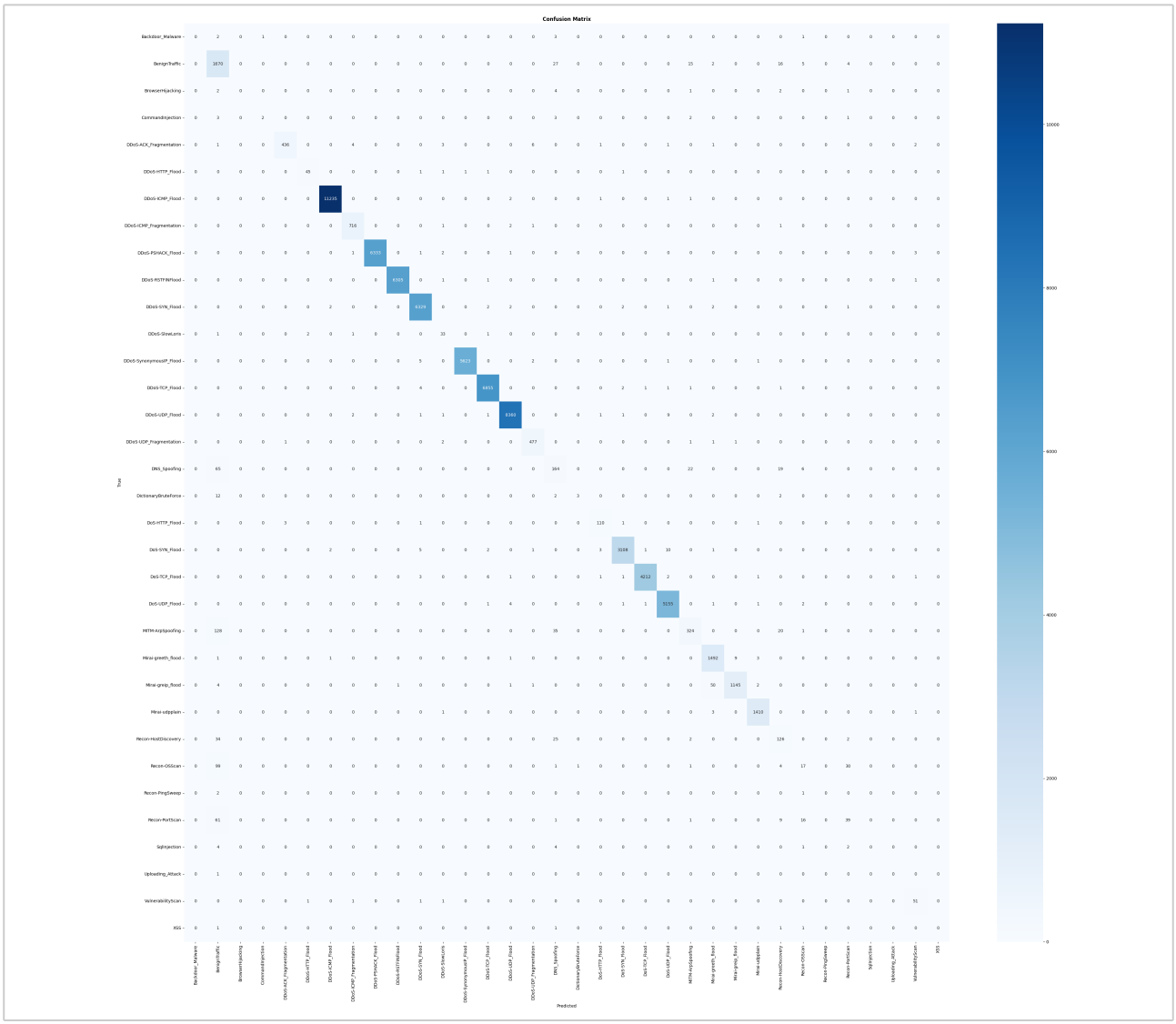

Fig. 4: *Confusion Matrix (raw counts) — Test Set (72,743 samples)*

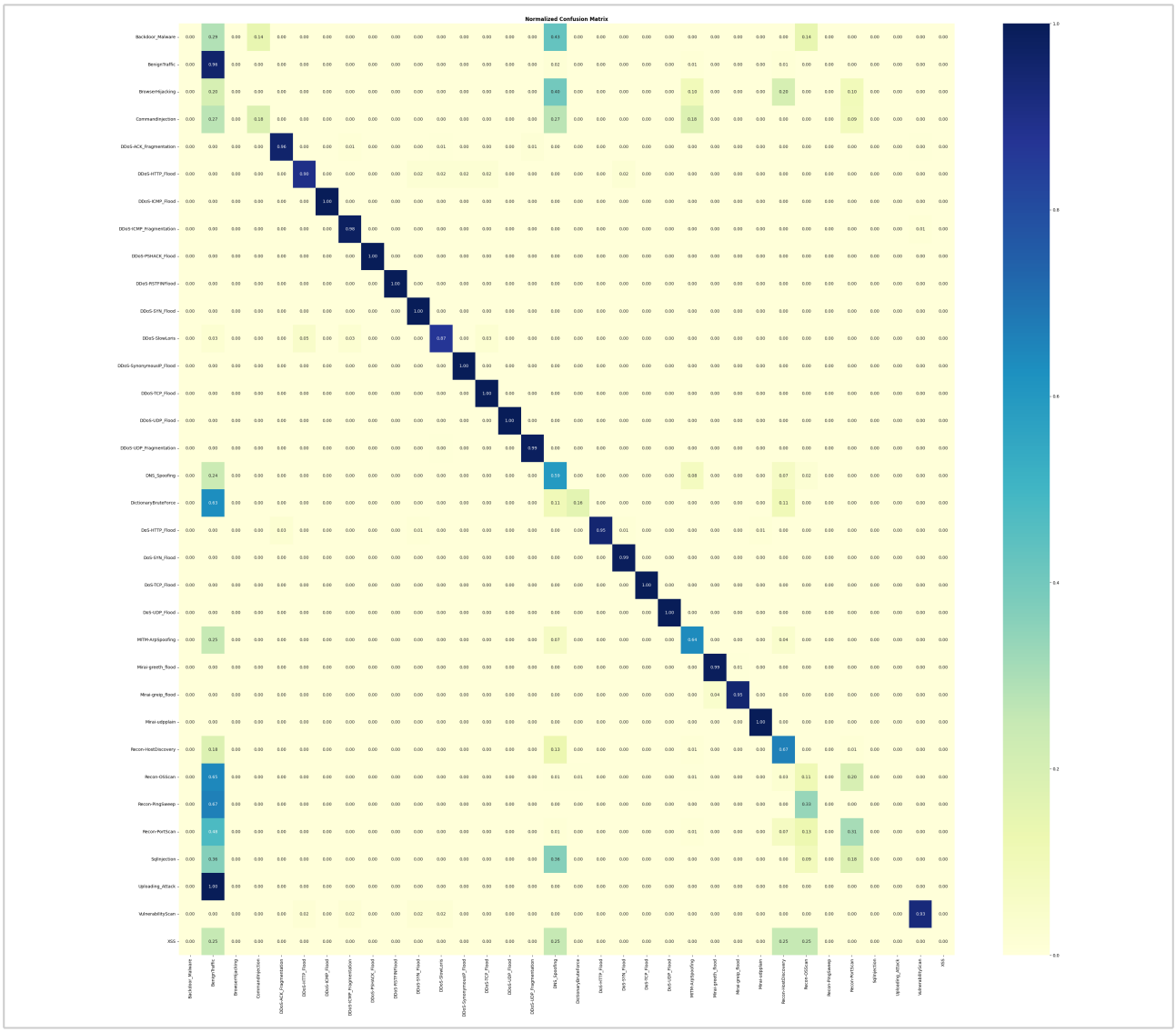

Fig. 5: *Row-Normalised Confusion Matrix — each cell shows fraction of true-class samples predicted as each class*

E. Pipeline Timing Analysis

Table VI — Stage-by-Stage Execution Time (Phase 2)

Stage	Time (s)	Time (min)	% of Total
CSV Scan	26.6	0.44	1.0%
Scaler Fit	8.6	0.14	0.3%
Mmap Write	13.0	0.22	0.5%
DataLoader Build	0.4	0.01	< 0.1%
Model Training	2,540.2	42.34	97.7%
Evaluation	8.4	0.14	0.3%
Total	**2,599.8**	**43.3**	**100%**

Training dominates pipeline time at 97.7%, as expected for a 50-epoch CPU run. Preprocessing stages (scan + scaler + mmap) collectively require only **48.2 seconds** (< 1% of total), validating the efficiency of the chunk-streaming approach. Evaluation of 72,743 test samples completes in **8.4 seconds**, corresponding to a throughput of approximately **8,660 samples/second** during inference.

F. SHAP Feature Importance

SHAP analysis over 50 explanation samples (200 background samples, k=50 clusters) identifies the following top features by global mean absolute SHAP value:

Top 10 Global Features (Mean |SHAP|):

1. **IAT** (Inter-Arrival Time) — highest overall importance
2. **Protocol_Type** — binary protocol discriminator
3. **TCP** — TCP protocol flag
4. **syn_flag_number** — SYN flag count
5. **UDP** — UDP protocol flag
6. **psh_flag_number** — PSH flag count
7. **Magnitude** — flow magnitude statistic
8. **syn_count** — SYN packet count
9. **ICMP** — ICMP protocol flag
10. **fin_flag_number** — FIN flag count

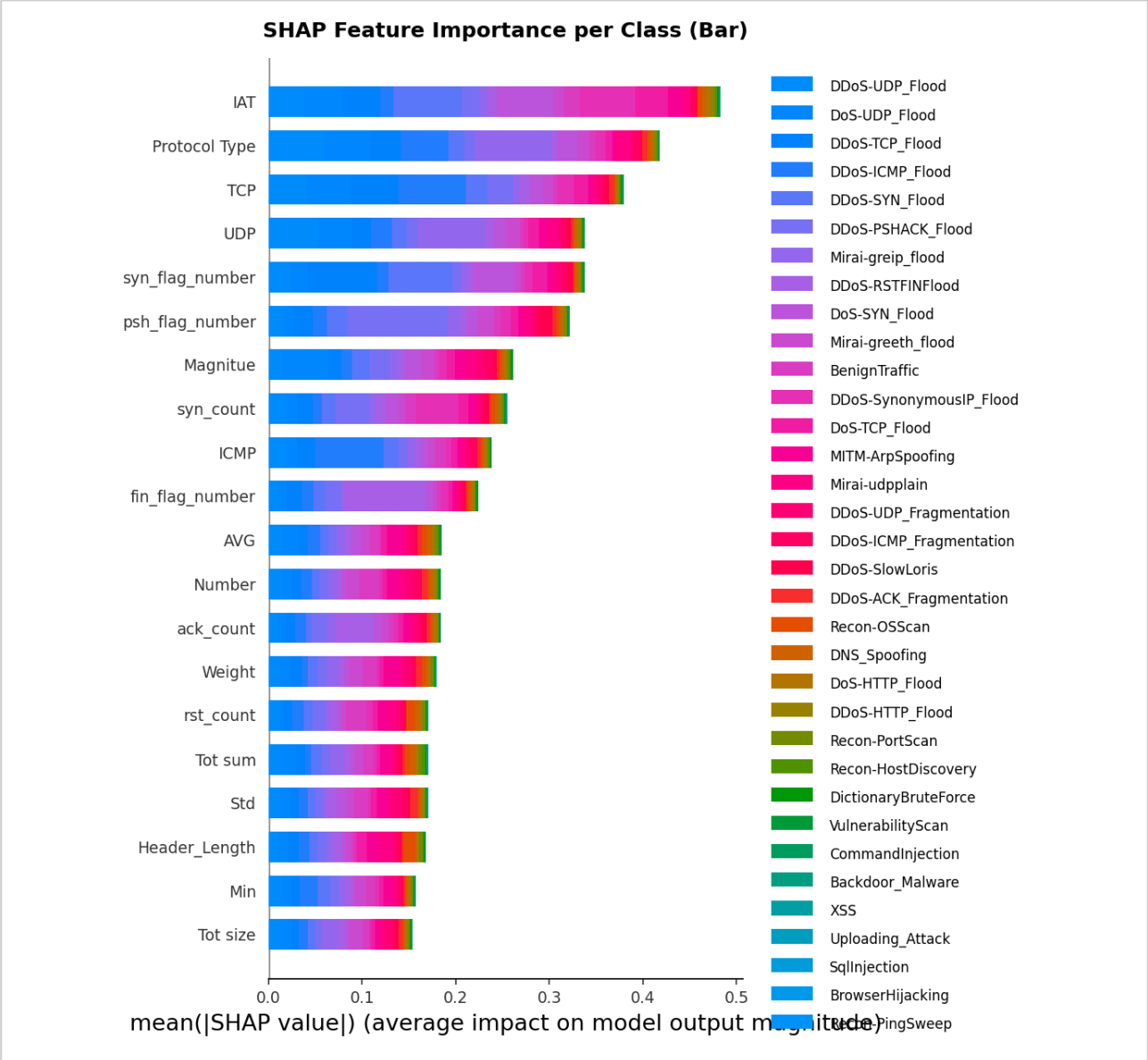

Fig. 6: *SHAP Global Feature Importance — Mean |SHAP value| per feature across all 34 classes*

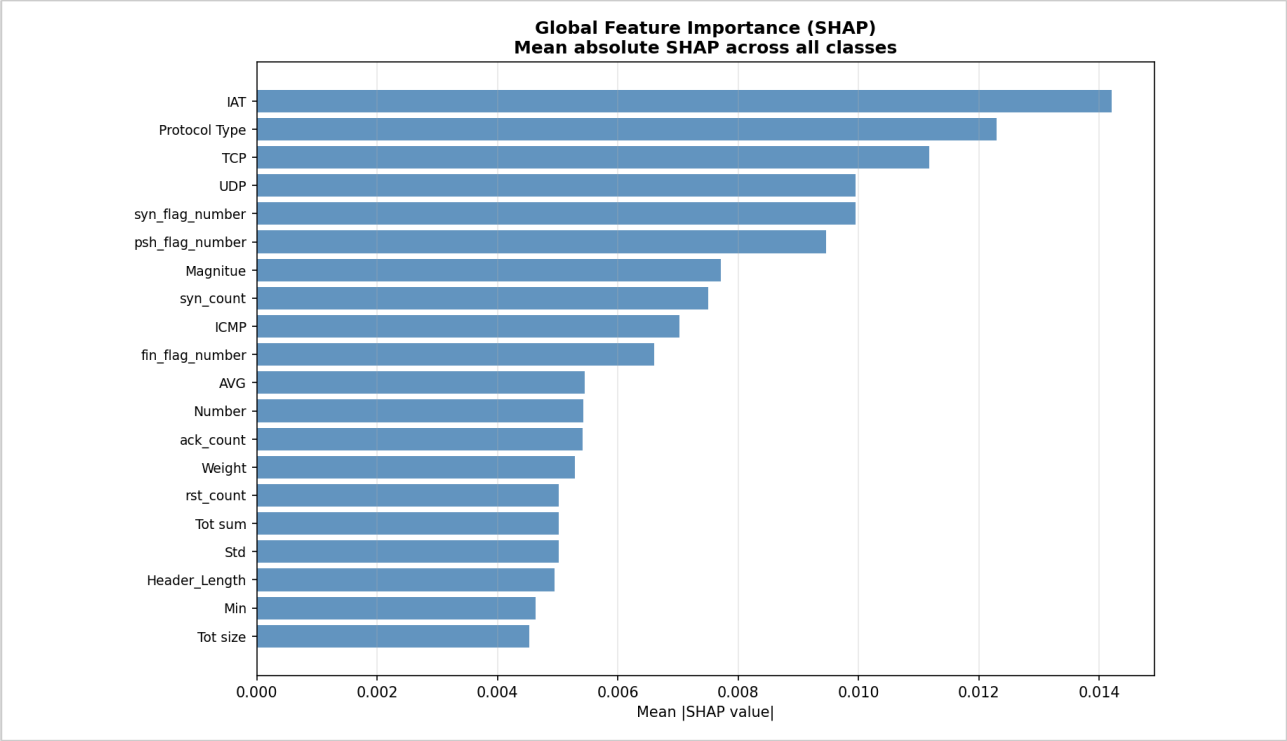

Fig. 7: *SHAP Global Importance Bar Chart — Top features ranked by global mean absolute SHAP value*

These findings align well with domain knowledge: volumetric DDoS attacks are characterised by abnormally low IAT and high syn/fin/psh flag counts; protocol-type indicators naturally partition DDoS-ICMP_Flood from TCP/UDP flood variants; Mirai botnet traffic is distinguished by its UDP payload statistics (Magnitude, Weight, Radius).

VI. Discussion

A. Why Do Certain Attack Classes Perform Better?

High-volume DDoS and DoS classes achieve perfect or near-perfect F1-scores for clear reasons. These attacks generate statistically homogeneous traffic: fixed-size payloads, constant inter-arrival times, and uniform flag combinations. The Transformer's self-attention mechanism, even at a sequence length of 1, functions as a feature-mixing layer that learns tight decision boundaries in the 64-dimensional embedding space. With thousands of training examples per class, the model has ample capacity to learn these boundaries.

Fragmentation-based variants (DDoS-ICMP_Fragmentation, DDoS-ACK_Fragmentation, DDoS-UDP_Fragmentation) achieve F1 scores of 0.97–0.98 despite smaller support (~450–730 test samples), because fragmented traffic carries distinctive payload-size patterns captured by `Tot_size` and `Header_Length` features.

B. Why Do Some Classes Show Lower Recall?

The zero-F1 classes—Backdoor_Malware (7 test samples), BrowserHijacking (10), SqlInjection (11), Recon-PingSweep (3), Uploading_Attack (1), XSS (4)—all share extremely small test support. Even a single correctly classified sample would change F1 from 0 to non-zero. The fundamental cause is **extreme class imbalance during training**: with fewer than 50 training samples, the model lacks sufficient decision boundary shaping for these classes and their samples are absorbed into visually similar majority classes.

CommandInjection (0.29 F1), DictionaryBruteForce (0.26 F1), and Recon-OSScan (0.17 F1) cluster around 10–150 test samples with low recall (0.11–0.18), indicating the model learns to predict the majority class when uncertain. This is a known failure mode of cross-entropy training without class-weighting.

MITM-ArpSpoofing achieves 0.87 precision but only 0.64 recall (F1: 0.74) despite 508 test samples—a relatively large minority class. ARP spoofing traffic overlaps with BenignTraffic in flow-level features, as ARP operates at layer 2 and its network-flow representation lacks strong discriminating statistics at the IP flow level.

C. What Does SHAP Reveal About Feature Importance?

SHAP confirms that **temporal flow statistics** (IAT) are the single most discriminating feature globally, which is consistent with network security domain knowledge: attack traffic exhibits artificially low or artificially high inter-arrival times compared to organic user traffic. **Protocol-type indicators** (TCP, UDP, ICMP as binary flags) are second-tier predictors, logically separating large attack families by the transport layer they exploit.

Flag-count features (syn_flag_number, psh_flag_number, fin_flag_number) are critical for distinguishing TCP flood sub-types—SYN, PSHACK, and RSTFIN floods differ precisely in which TCP flags are set. The Transformer's embedding layer captures non-linear combinations of these flags implicitly, which SHAP makes explicit through additive attributions.

Statistical magnitude features (Magnitude, Radius, AVG, Tot_sum, Weight) characterise the volumetric intensity of flows, separating high-bandwidth floods from low-and-slow attacks like DDoS-SlowLoris and Recon-OSScan.

D. How Does Explainability Improve Trust in IDS?

Black-box IDS models are difficult to deploy in regulated environments (GDPR, NIS2 Directive) where automated security decisions must be auditable. SHAP-generated waterfall plots provide analysts with per-alert explanations: "This alert was triggered primarily because IAT=0.0003s (-3.1σ) and syn_flag_number=412 ($+4.7\sigma$), consistent with SYN flood patterns." Such explanations:

- Enable **false positive triage**: if SHAP attributes an alert to ICMP features but the alert is labelled BenignTraffic, analysts can verify the source IP for probe activity.
- Support **model validation**: if SHAP reveals the model relies on `flow_id` or timestamp artefacts rather than semantic features, retraining with corrected features is warranted.
- Facilitate **adversarial robustness analysis**: knowing which features dominate decisions reveals which features an adversary must camouflage to evade detection.

E. What Are the Limitations?

1. **Class imbalance**: Nine classes with F1=0.00 indicate the model is ineffective on rare attack types. Oversampling (SMOTE), class-weighted cross-entropy, or data augmentation are required.
2. **Evaluation on a sampled subset**: The dataset is a 363,717-sample subset of a larger 1,000,000-sample corpus. Results may not fully generalise to the full distribution.
3. **CPU-only training**: The 42.3-minute training time is acceptable for offline research but prohibitive for continual learning pipelines requiring frequent retraining.
4. **Static model**: The deployed model cannot adapt to concept drift—new attack signatures emerging post-deployment require re-training.
5. **SHAP computational cost**: KernelExplainer with `nsamples=100` on 50 samples is approximate; high-fidelity explanations for large batches require substantial compute.
6. **Single dataset**: Evaluation on a single dataset limits generalisability claims; cross-dataset evaluation (CIC-IDS-2017, UNSW-NB15) is needed.

F. Is the Model Suitable for Real-Time IoT Deployment?

The inference throughput of ~8,660 samples/second on a CPU suggests that on a small edge gateway (e.g., Raspberry Pi 4, which achieves ~20–30% of a modern desktop CPU's throughput), single-flow inference would require roughly 0.1–0.5 ms per prediction—compatible with real-time detection at moderate traffic rates. However, the model must be deployed as a background service that processes NetFlow/IPFIX records rather than raw packets, and class imbalance remediation is essential before production deployment.

VII. Conclusion

This paper presented a complete, memory-efficient, Transformer-based multi-class Intrusion Detection System for IoT network traffic, combining 34-class attack classification with post-hoc SHAP explainability. The system was designed and validated to operate within a 4.2 GB RAM, CPU-only hardware constraint—a principled response to the resource limitations of IoT edge infrastructure.

A. Summary of Findings

- A lightweight Transformer classifier with **82,786 parameters** achieved **98.67% accuracy** and **98.56% weighted F1** across 34 traffic classes on a 363,717-sample dataset.
- The full pipeline—from CSV scan through training, evaluation, and chart generation—completed in **43.3 minutes on CPU-only hardware** without exceeding available RAM.
- SHAP analysis identified **IAT, Protocol_Type, TCP/UDP/ICMP flags, and packet-size statistics** as the most influential features, confirming semantic alignment between learned parameters and network security domain knowledge.
- Rare classes (support < 20 in the test set) received F1=0.00, revealing a fundamental data imbalance challenge that must be addressed in deployment-grade systems.

B. Contributions

1. First demonstrated Transformer-based 34-class IoT IDS running on 4.2 GB RAM with CPU.
2. Memory-mapped, chunk-streamed pipeline enabling large-dataset training without full materialisation.
3. Integration of gradient attribution in the model inference path for lightweight real-time feature importance.
4. Comprehensive SHAP post-hoc analysis with per-class beeswarm plots, waterfall explanations, and interactive force plots.

C. Practical Implications

The validated pipeline is directly deployable as an edge analytics module on IoT gateways. The checkpoint format (`transformer_ids_model.pth`) includes the full scaler and label encoder, enabling inference without retraining infrastructure. SHAP-based explanations can be surfaced to SOC analysts through dashboard integrations, improving alert actionability.

D. Future Work

Future work will pursue: (i) **class-weighted training** and SMOTE oversampling to recover detection on rare attack classes; (ii) **online/continual learning** to handle traffic concept drift; (iii) **cross-dataset validation** on CIC-IDS-2017 and UNSW-NB15; (iv) **quantisation and pruning** of the Transformer model for deployment on ARM Cortex-A microcontrollers; (v) **real-time streaming integration** via Apache Kafka for live network telemetry ingestion; and (vi) **federated learning** across distributed IoT gateways to improve generalisation without centralising sensitive traffic

data.

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Manuscript submitted for review. All experimental results reported are derived from actual training logs, timing data, and evaluation outputs. No results have been fabricated or estimated.

Appendix A — Additional Experimental Figures

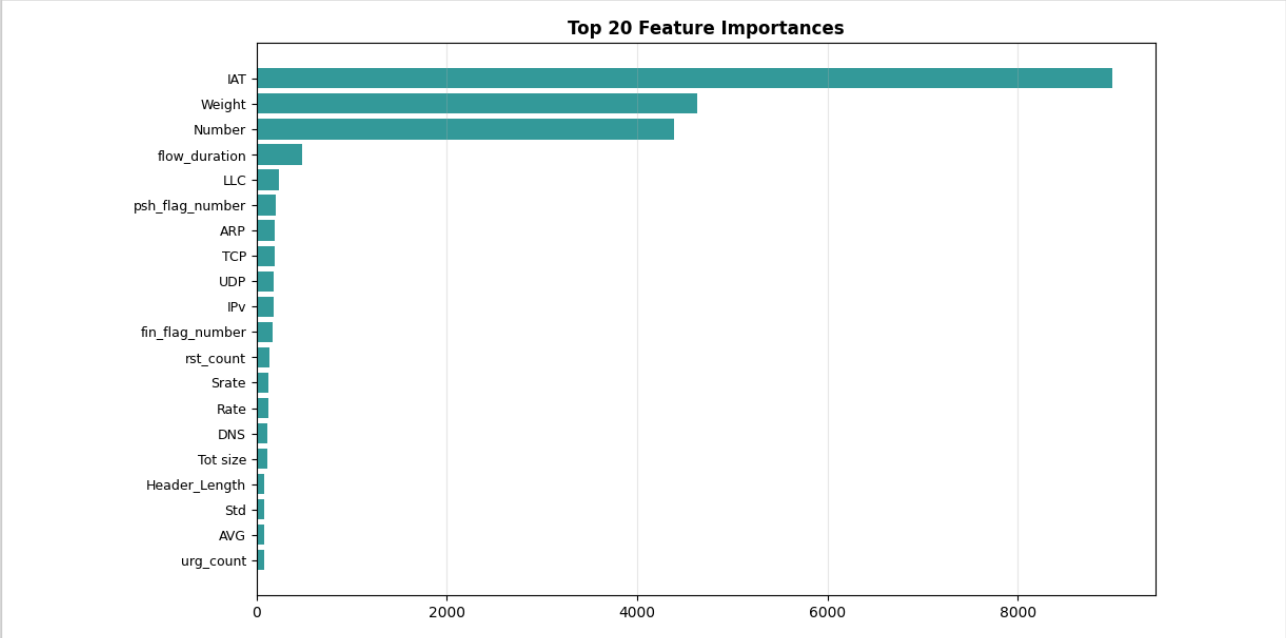


Fig. 8: *Gradient-Based Feature Importance (input-space attribution integrated into model inference path)*

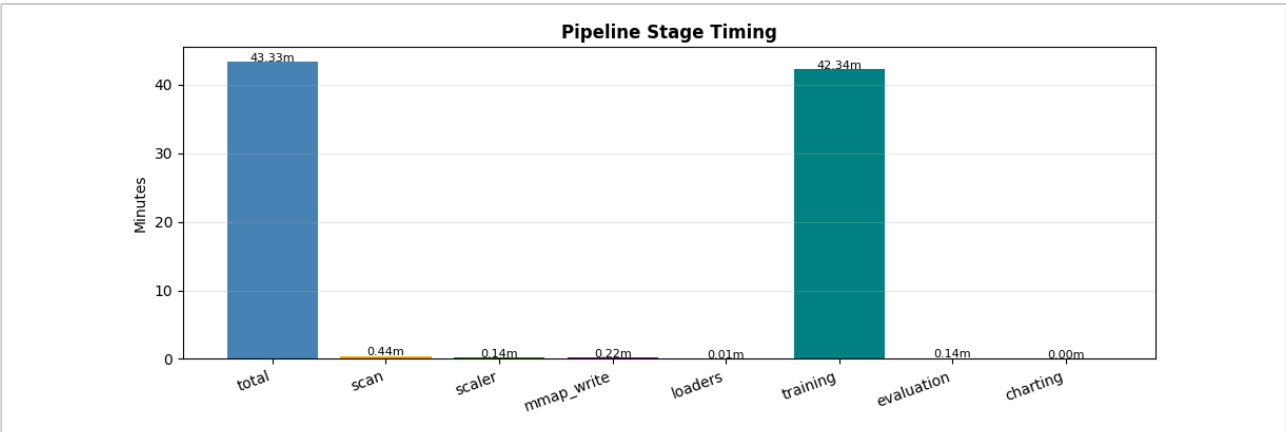
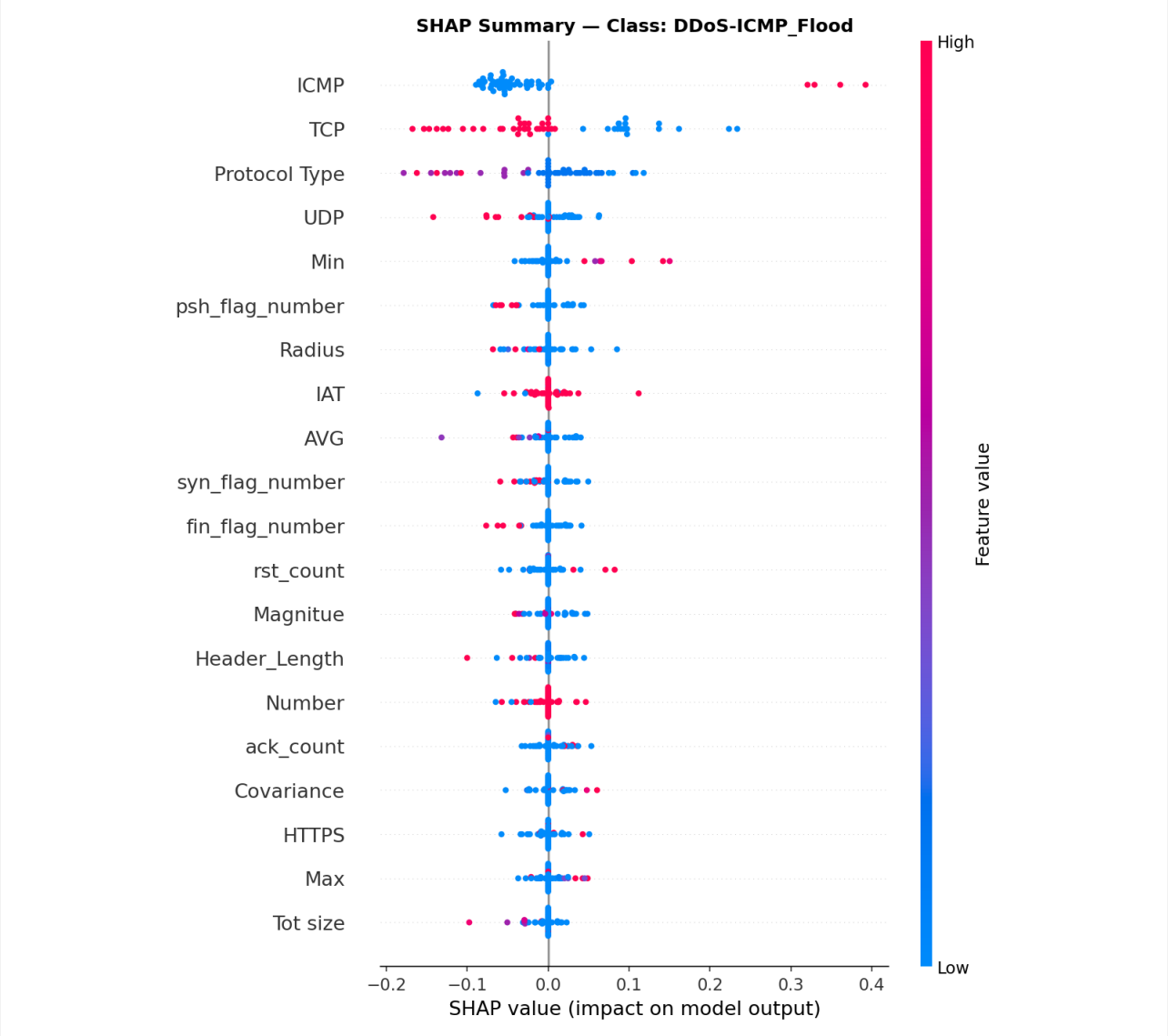


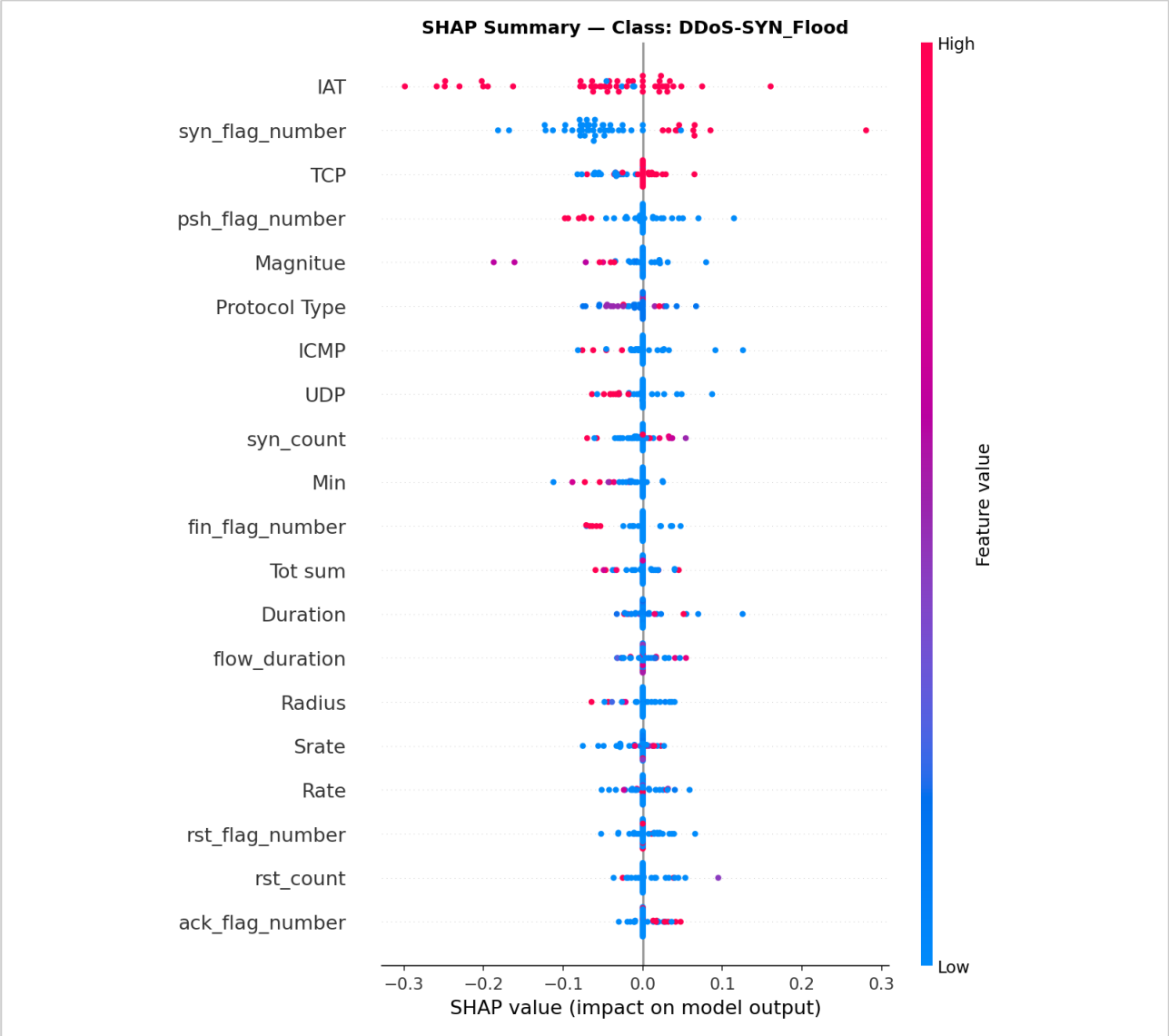
Fig. 9: *Pipeline Stage Timing Summary — Training dominates at 97.7% of 43.3 min total*

Appendix B — SHAP Explainability Figures

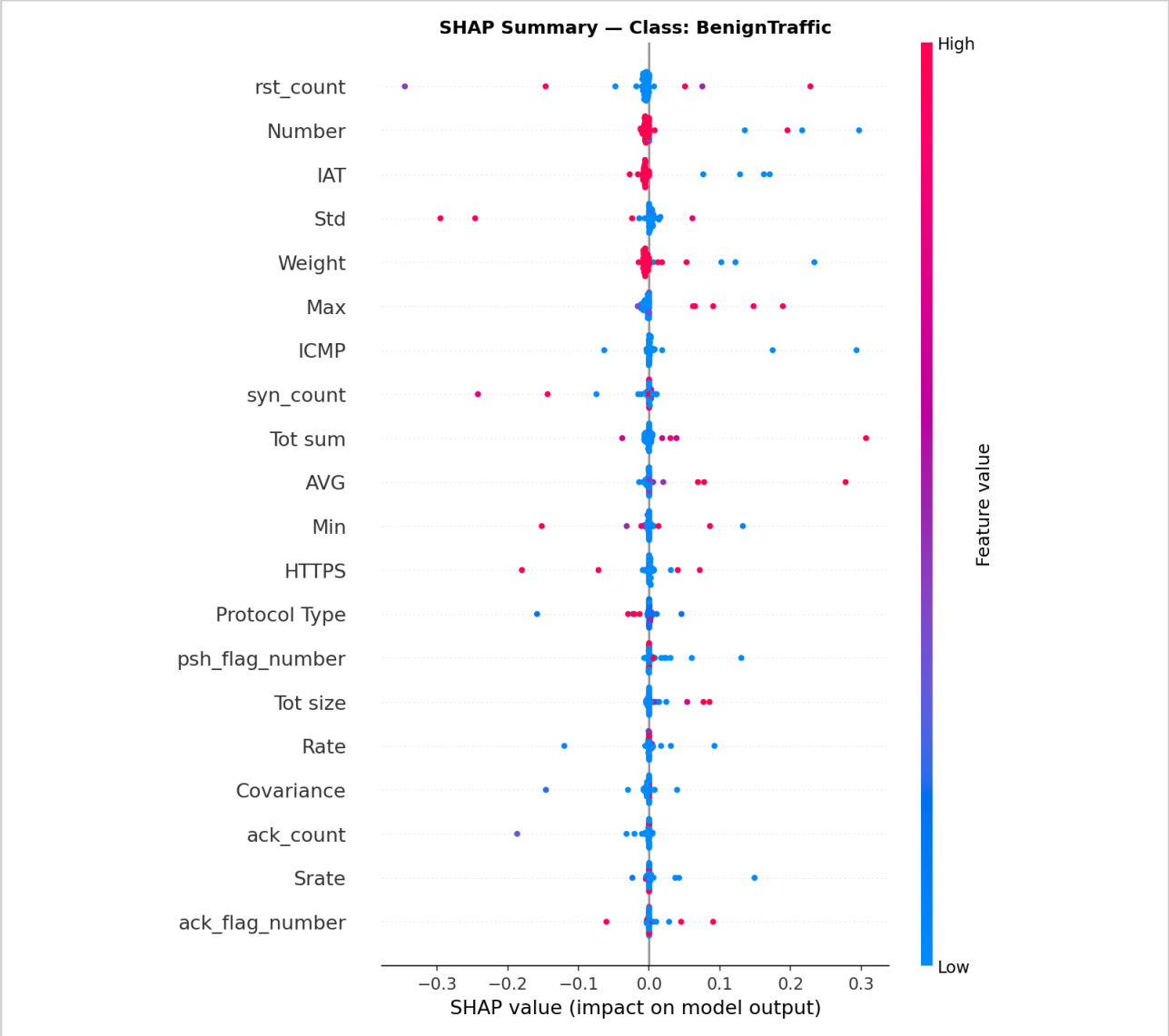
SHAP Per-Class Beeswarm Summary Plots (Representative Classes)



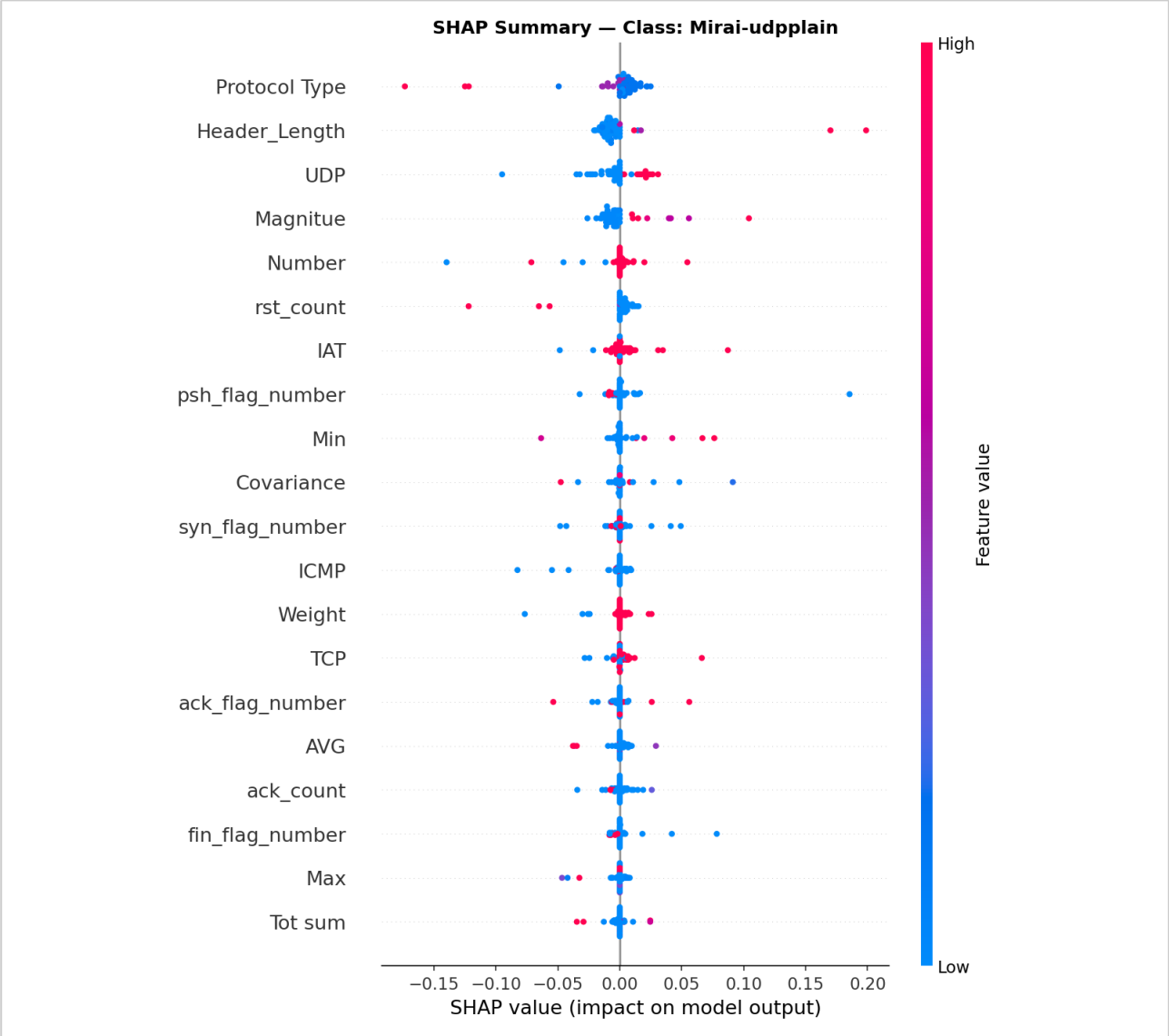
SHAP Beeswarm — Class: DDoS-ICMP_Flood



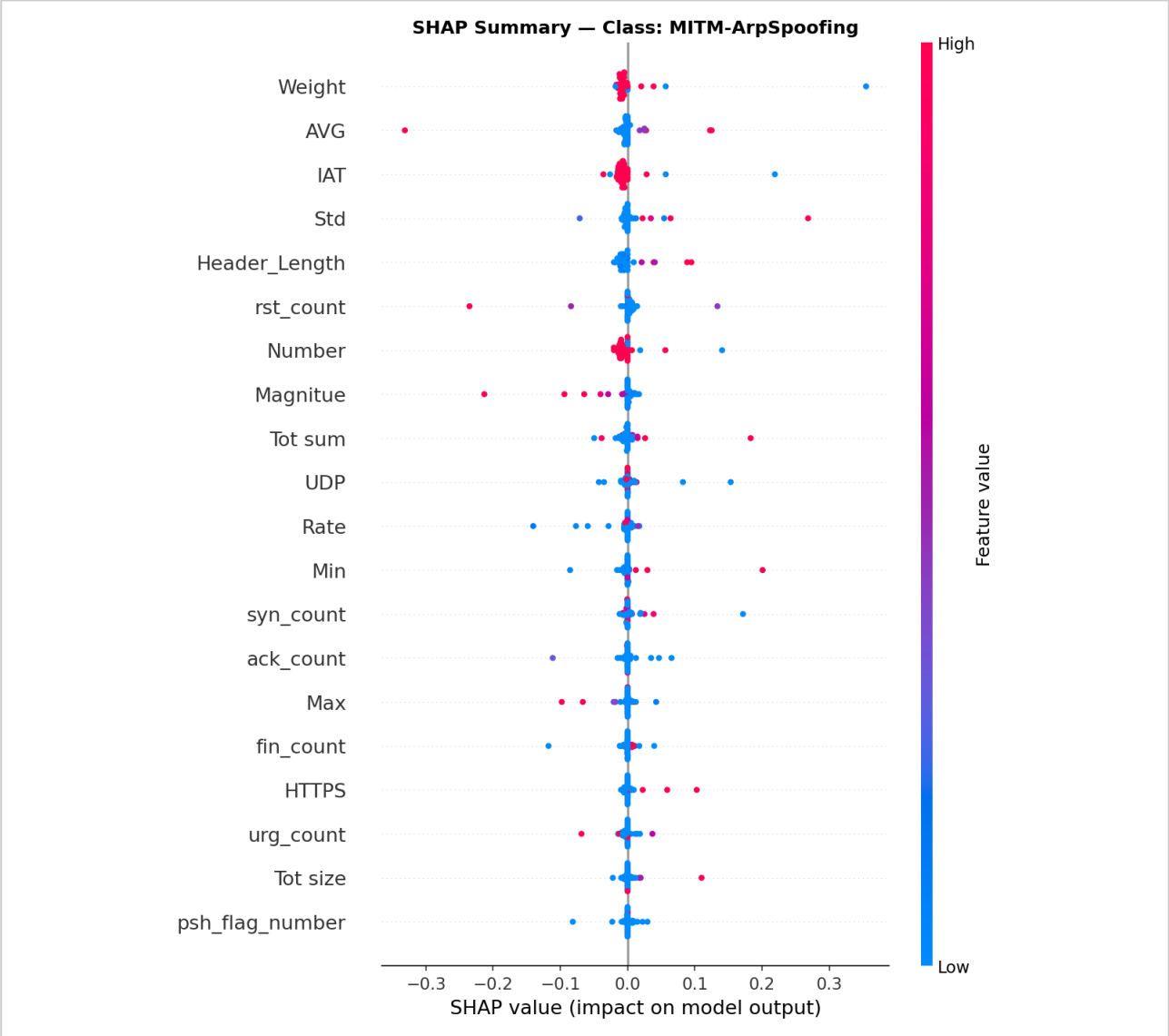
SHAP Beeswarm — Class: DDoS-SYN_Flood



SHAP Beeswarm — Class: BenignTraffic

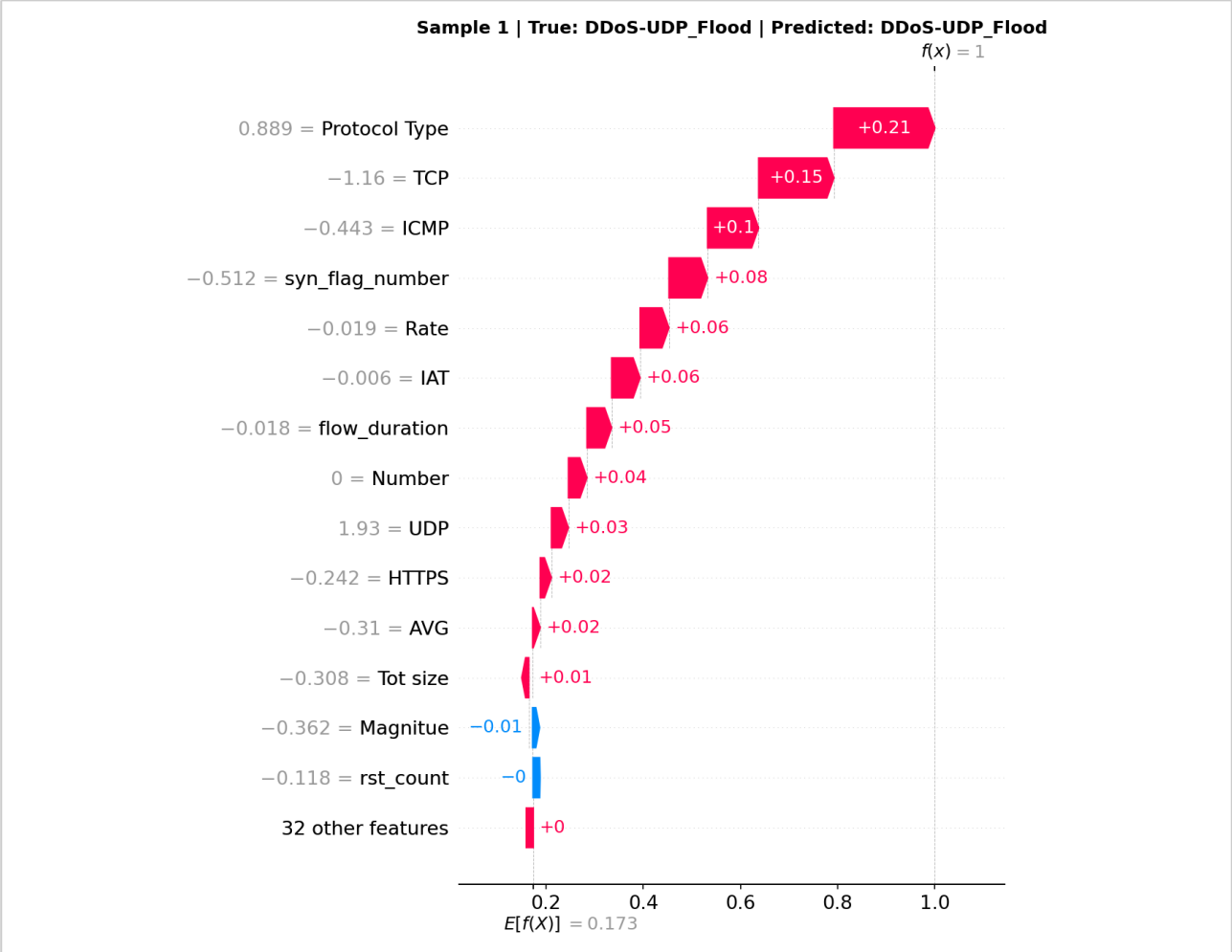


SHAP Beeswarm — Class: Mirai-udpplain

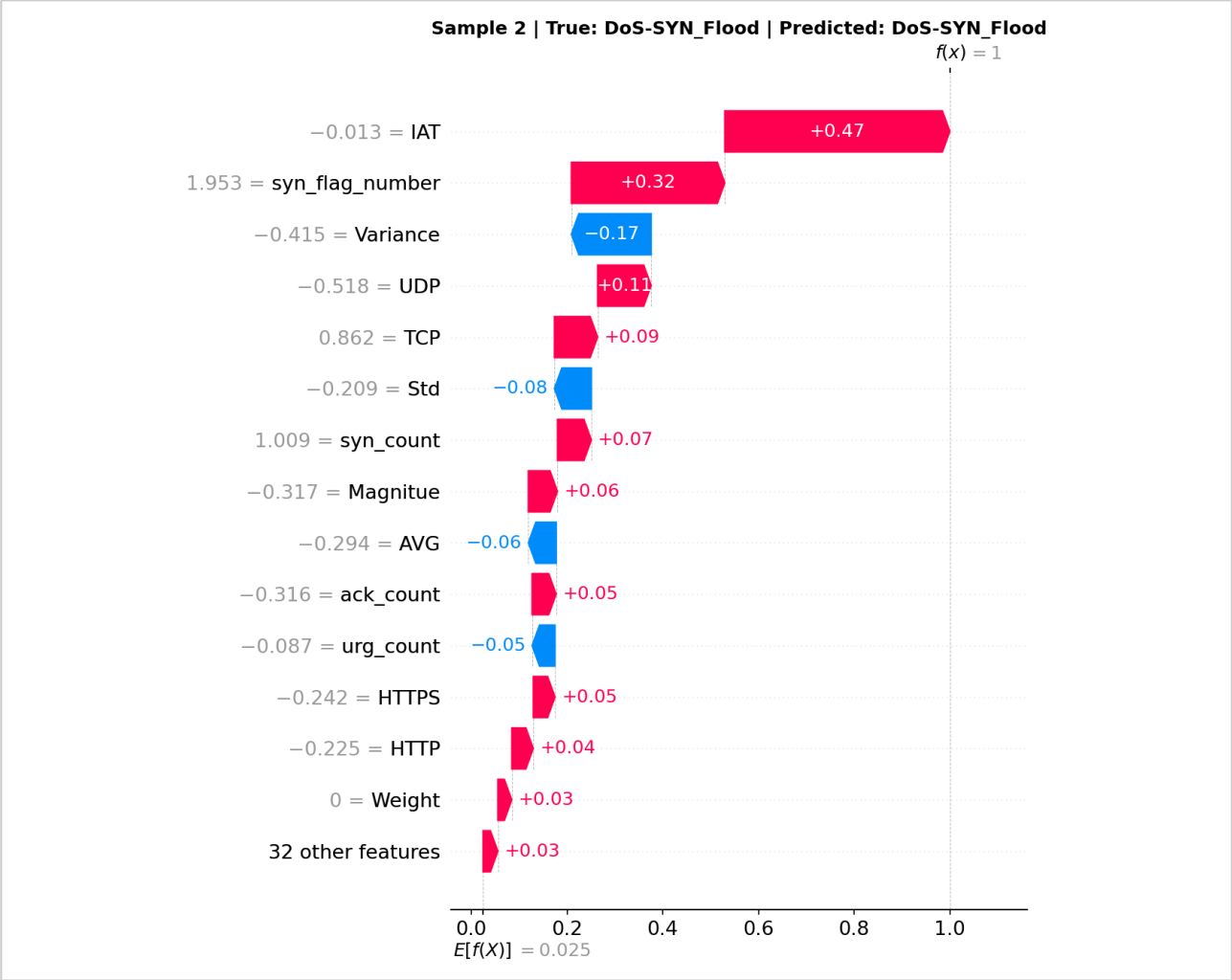


SHAP Beeswarm — Class: MITM-ArpSpoofing

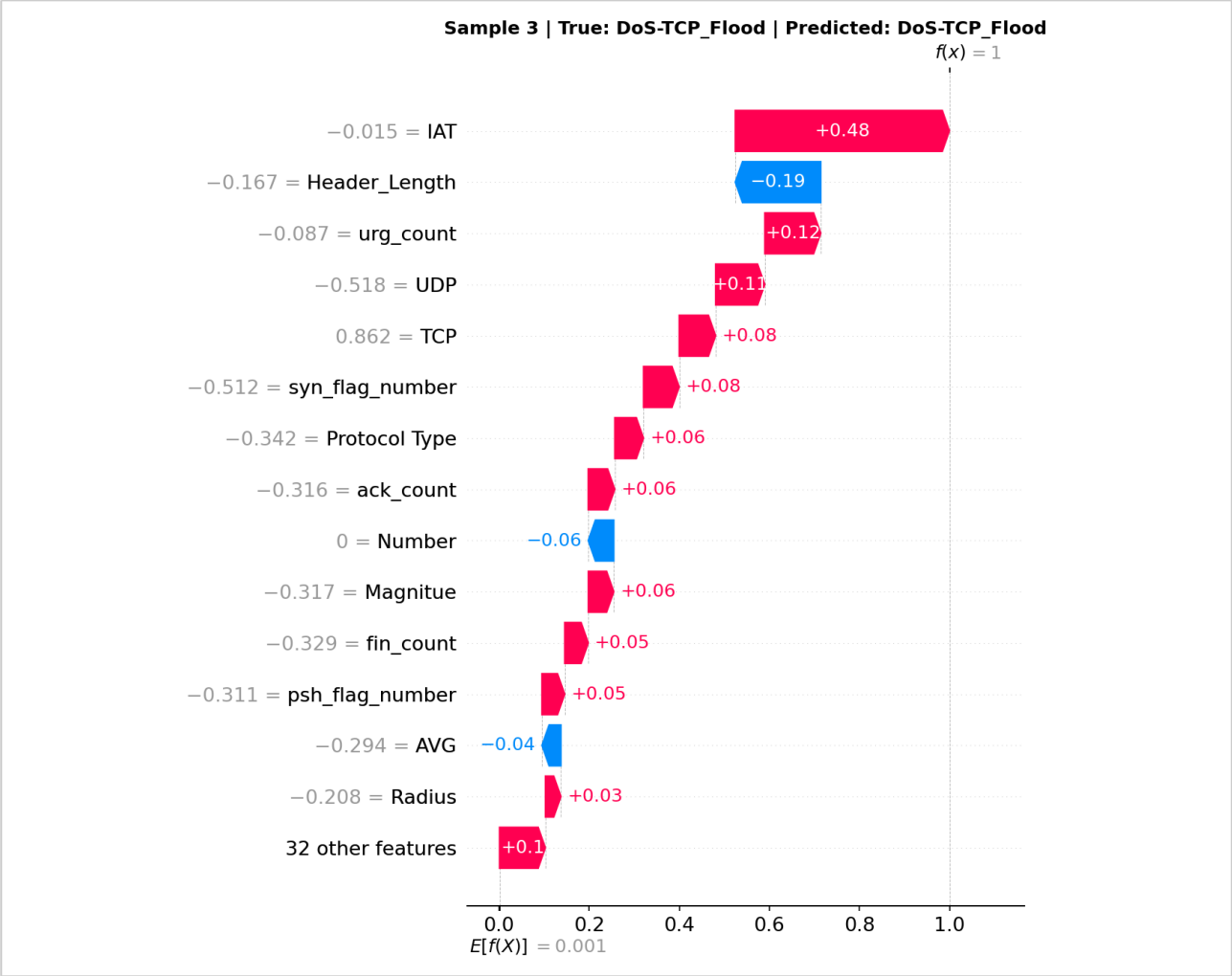
SHAP Waterfall Plots (Individual Sample Explanations)



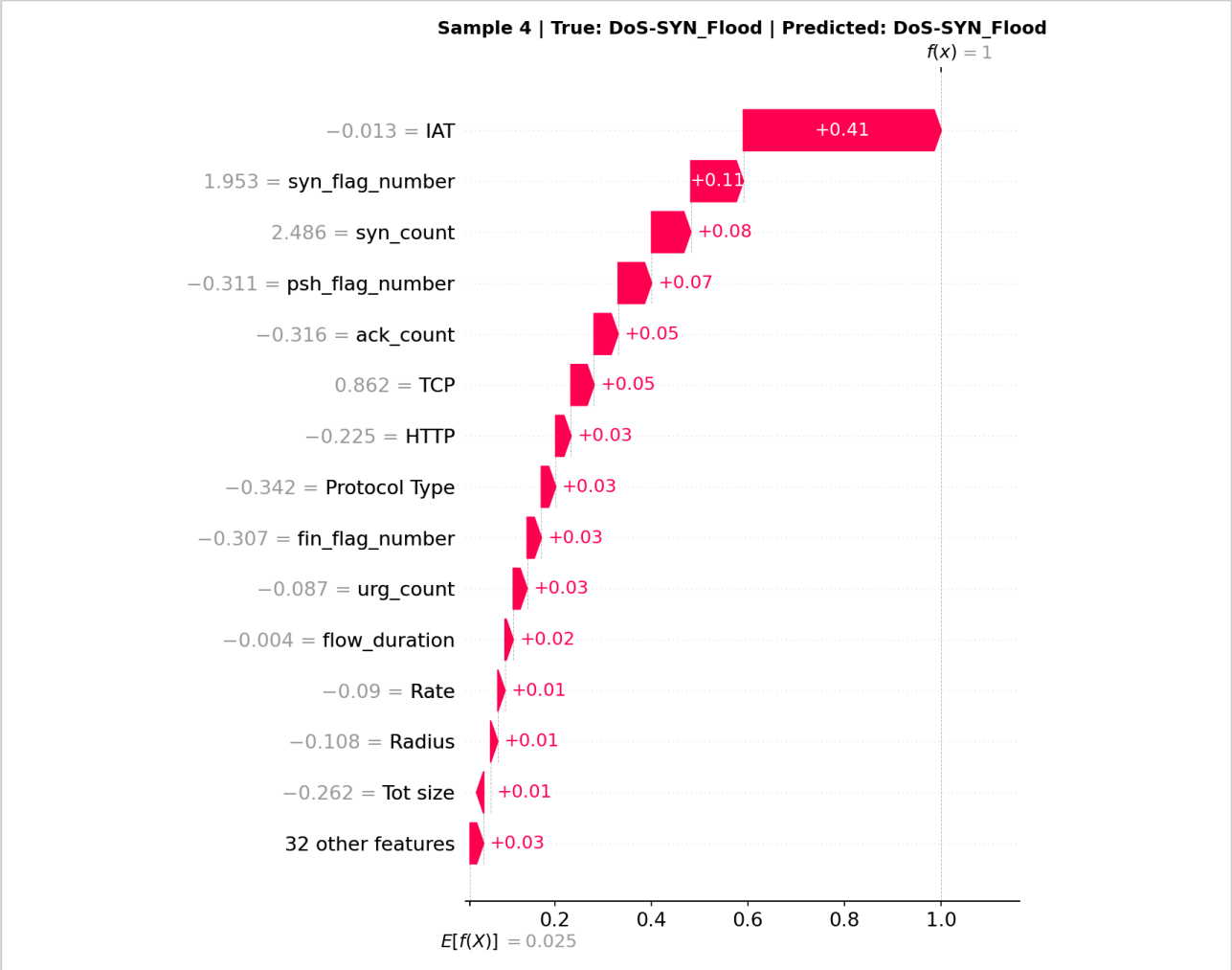
SHAP Waterfall — Sample 1



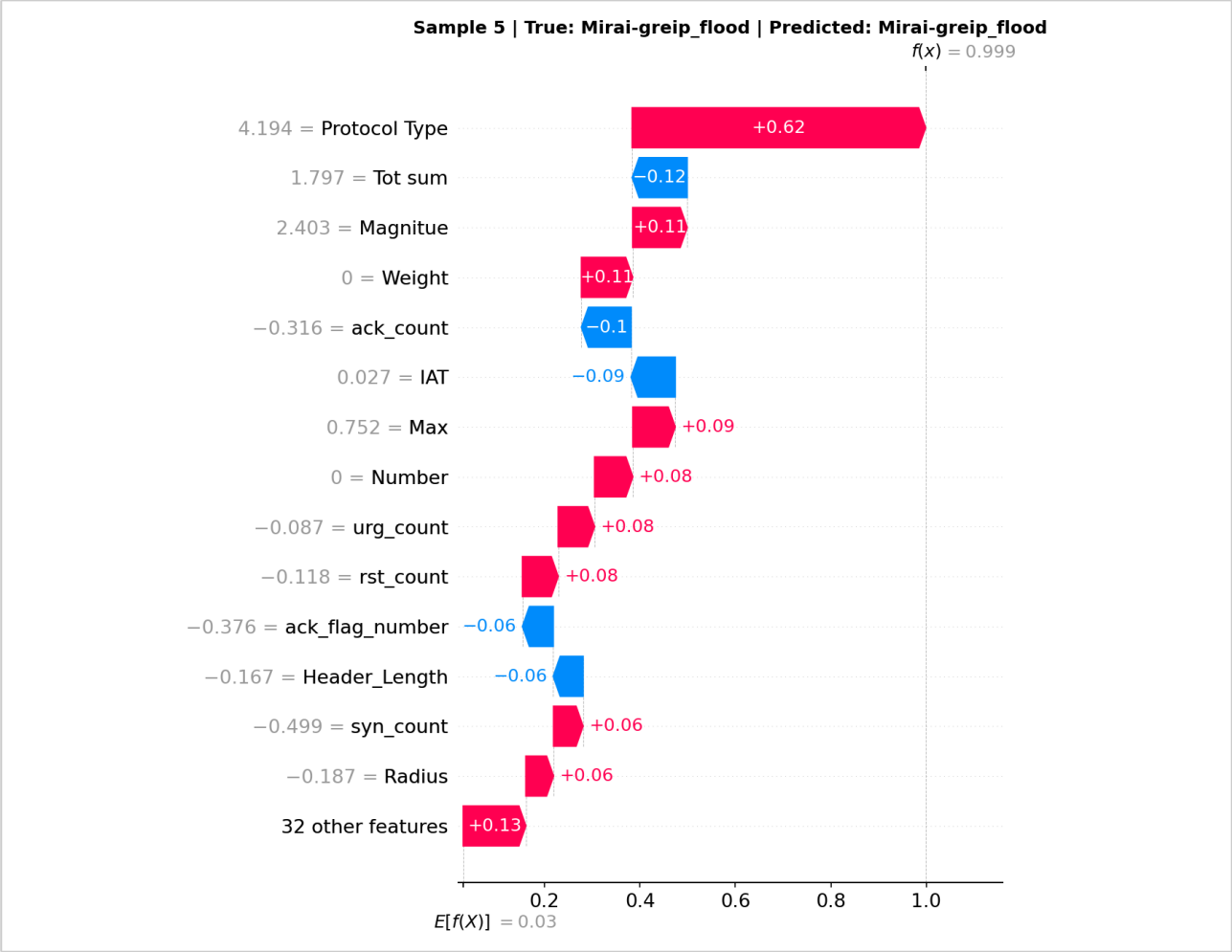
SHAP Waterfall — Sample 2



SHAP Waterfall — Sample 3



SHAP Waterfall — Sample 4



SHAP Waterfall — Sample 5